

MedTrack Hospital Analytics

Module 8: Documentation and Project Delivery

Hospital Operations & Patient Analytics Dashboard

Project Status	Completed	Visualization Tool	Tableau Desktop / Public
Dashboards	4 integrated dashboards	Data Processing	Python / Pandas / NumPy
Core KPIs	4 headline KPIs + supporting measures	Delivery Format	Tableau workbook + documentation

Purpose: This document records the data source, KPI logic, dashboard design, healthcare operations methodology, project structure, delivery approach, and final dashboard outputs for the MedTrack Hospital Analytics project.

1. Project Overview

MedTrack Hospital Analytics is an interactive Tableau-based analytics solution designed to convert hospital admission and operational data into practical management insights. The project focuses on patient volume, revenue, occupancy, length of stay, readmissions, department performance, bed capacity, and staff/resource utilization.

The final solution contains four connected dashboard views. Together they support hospital-level monitoring, patient-flow analysis, department comparison, and resource-capacity planning.

Project objectives

- Create a clean and analysis-ready hospital dataset.
- Engineer meaningful KPIs for hospital operations.
- Build interactive Tableau dashboards with filters and dashboard interactions.
- Compare departments using patient, financial, capacity, and efficiency measures.
- Present the completed work in a portfolio-ready documentation package.

2. Dataset Sources and Data Preparation

The project uses a hospital patient-admission dataset containing operational records such as admission information, department, admission type, dates, length of stay, revenue, occupancy/resource measures, readmission status, and staff/capacity attributes. The dataset supplied for the project was prepared for analysis before being connected to the Tableau workbook.

Data preparation workflow

- Reviewed fields and data types before analysis.
- Checked date fields for valid admission and discharge dates.
- Validated numeric measures such as revenue, length of stay, occupancy, beds, doctors, and nurses.
- Checked categorical values such as department and admission type for consistency.
- Handled missing/invalid records where required and validated the final analytical dataset.
- Connected the prepared data to Tableau for calculated fields, visualizations, filters, and dashboard actions.

Primary analytical dimensions

Dimension	Examples / Use
Department	Cardiology, Emergency, General Medicine, Gynecology, Neurology, Oncology, Orthopedics, Pediatrics
Admission Type	Planned, Emergency, Referral
Date	Admission month/date and discharge date for trend analysis
Patient Flow	Admissions, discharge activity, readmission status
Operations	Revenue, occupancy, length of stay, beds, staff and resource availability

3. KPI Definitions and Business Logic

The dashboard headline KPIs provide a quick hospital-wide view. Supporting charts then break these measures down by department, month, admission type, and resource category.

KPI	Definition	Dashboard Use
Total Admissions	Count of valid patient admission records / Admission IDs.	Measures overall patient volume.
Total Revenue	Sum of recorded revenue across the selected dataset/filter context.	Shows the financial value generated by patient activity.
Average Length of Stay	Average number of days patients stayed in the hospital.	Supports bed planning and operational efficiency.
Average Occupancy	Average occupied-bed rate expressed as a percentage.	Shows how heavily hospital capacity is being used.
Readmissions	Count/sum of records identified as readmission cases.	Helps identify repeat patient utilization.
Bed Utilization	Average or calculated utilization of available bed capacity.	Supports capacity and resource planning.
Department Efficiency	Comparative score used to rank department operational performance.	Highlights stronger and weaker departments.

Current dashboard snapshot	Value
Total Admissions	4,000
Total Revenue	₹29,73,72,500
Average Length of Stay	3.814 Days
Average Occupancy	71.95%

4. Dashboard Guide

The final workbook is organized into four main dashboard views. Each dashboard has a specific operational purpose while sharing common filters and KPI context.

Dashboard	Purpose	Key visuals / decisions
1. Hospital Overview & Analytics	High-level hospital performance monitoring.	Admissions, revenue, occupancy, length of stay, revenue by department, occupancy by department, monthly admissions, readmission analysis, patient movement.
2. Patient Flow Dashboard	Understand how patients enter, move through, and leave the hospital.	Patient movement, discharge tracking, average stay, admission trends, peak patient load.
3. Department Analytics	Compare departments and identify operational differences.	Patient volume, efficiency score, readmissions, treatment capacity, total bed capacity.
4. Resource Utilization	Monitor utilization and resource availability.	Bed utilization, occupancy rate, staff allocation, bed capacity, resource availability.

Interactive functionality

- Department filters allow users to isolate one or more departments.
- Admission Type filters separate Planned, Emergency, and Referral activity.
- Admission Date / Month filters support time-based analysis.
- Dashboard actions and selections allow charts to be used together for interactive exploration.
- Measure/parameter controls are used where a dashboard requires switching between operational measures.

5. Healthcare Operations Methodology

The methodology follows a practical hospital-operations analytics cycle: collect and prepare data, define business measures, visualize performance, identify operational patterns, and support management decisions.

Stage	Method Applied
1. Data Collection	Use the supplied hospital admission and operations dataset as the analytical source.
2. Data Cleaning	Validate dates, numeric measures, categorical labels, missing values, and record consistency.
3. KPI Engineering	Create measures for admissions, revenue, occupancy, length of stay, readmissions, bed utilization, capacity and efficiency.
4. Trend Analysis	Analyze monthly admissions and discharge patterns to identify changes in patient load.
5. Department Analysis	Compare departments on volume, revenue, occupancy, readmissions, efficiency and capacity.
6. Resource Planning	Use beds, occupancy, doctors and nurses to support capacity and staffing decisions.
7. Decision Support	Use filters and visual comparisons to identify high-load periods, capacity pressure and department-level differences.

Example operational insights from the final dashboard

- Patient admissions are highest at the beginning of the displayed year and reduce substantially during the middle and later months, with a small recovery toward December.
- Occupancy rates remain close to the low-70% range across departments, indicating relatively consistent utilization.
- Department patient volumes differ, making department-level comparisons important for staffing and capacity decisions.
- General Medicine shows the largest displayed treatment-capacity value, while several specialist departments operate at lower capacity.
- Readmission counts vary by department and can be used to prioritize further clinical/operational investigation.

6. Project Structure and Delivery

The project is organized so that data, scripts, Tableau workbooks, and documentation can be maintained separately. This makes the work easier to review, reproduce, and present as a portfolio project.

Folder	Purpose
/scripts	Python scripts used for data preparation, cleaning, validation or supporting analysis.
/data	Source and cleaned hospital datasets used by the project.
/dashboard	Tableau workbook, packaged workbook and dashboard-related assets.
/docs	Module documentation, KPI definitions, methodology, screenshots and final reports.

Recommended delivery package

- GitHub repository containing the organized project folders.
- Final Tableau workbook (.twb or packaged .twbx, as applicable).
- Final Module 8 documentation PDF.
- Cleaned dataset or permitted project dataset.
- Optional Tableau Public link if the workbook is published.

Deployment / sharing

The project can be delivered through a GitHub repository for source/documentation management and Tableau Public for optional dashboard sharing. If the dataset contains restricted or sensitive information, only the permitted sample/derived data should be published.

7. Technology Stack and Evaluation

Area	Tools / Technologies
Data Collection	Python, Hospital Dataset, Patient Admission Dataset
Data Processing	Pandas, NumPy
Data Cleaning	Python
Visualization	Tableau Desktop / Tableau Public
Dashboard Integration	Tableau Filters, Parameters, Actions
Documentation	Markdown, GitHub, PDF documentation

Milestone	Focus Area	Metric	Target
1	Data Collection & Cleaning	Dataset Completeness	>95% Complete
2	KPI Engineering	KPI Accuracy	6+ KPIs Generated
3	Dashboard Development	Dashboard Functionality	4 Dashboards Integrated
4	Documentation & Delivery	Project Quality	Portfolio Ready

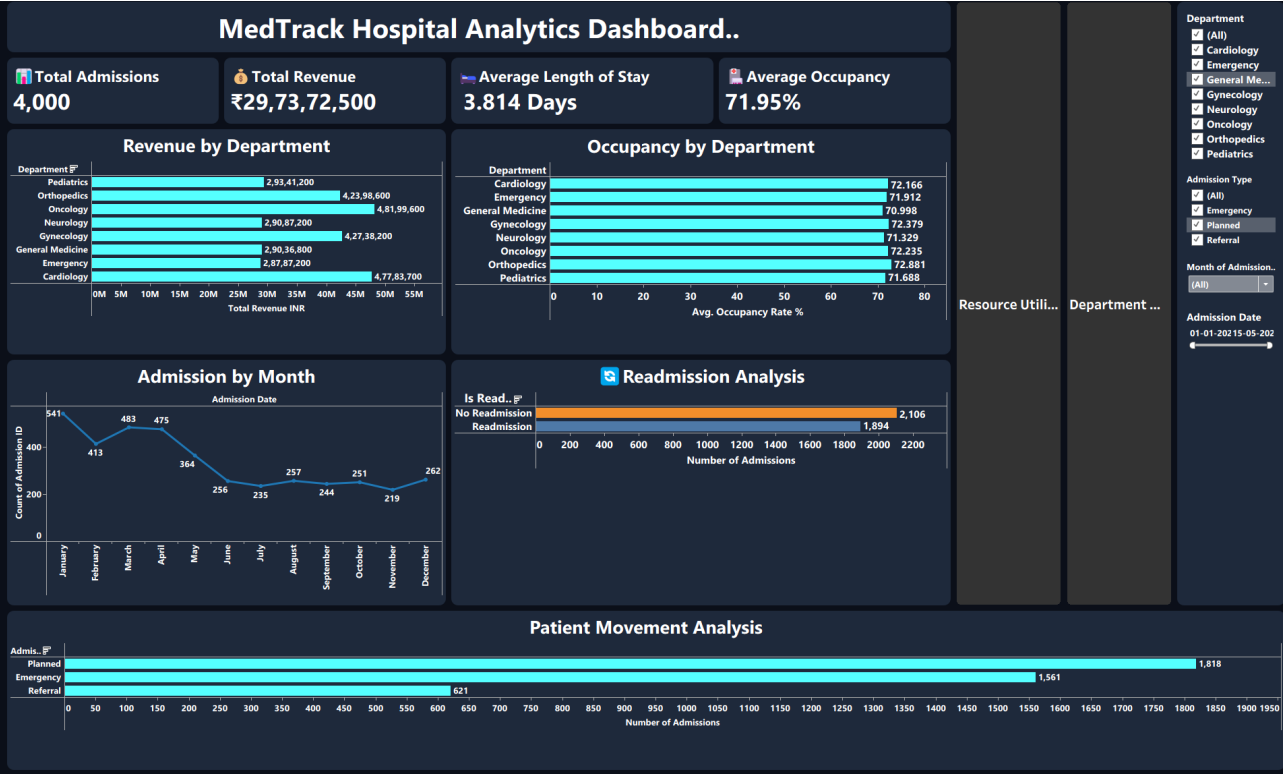
Completion statement

The MedTrack project meets the intended Module 8 delivery structure by documenting the data source, KPI definitions, dashboard purpose, healthcare operations methodology, project organization, technology stack, evaluation criteria, and final dashboard outputs. The four completed Tableau dashboards are included at the end of this document as evidence of the final deliverable.

8. Final Dashboard Evidence

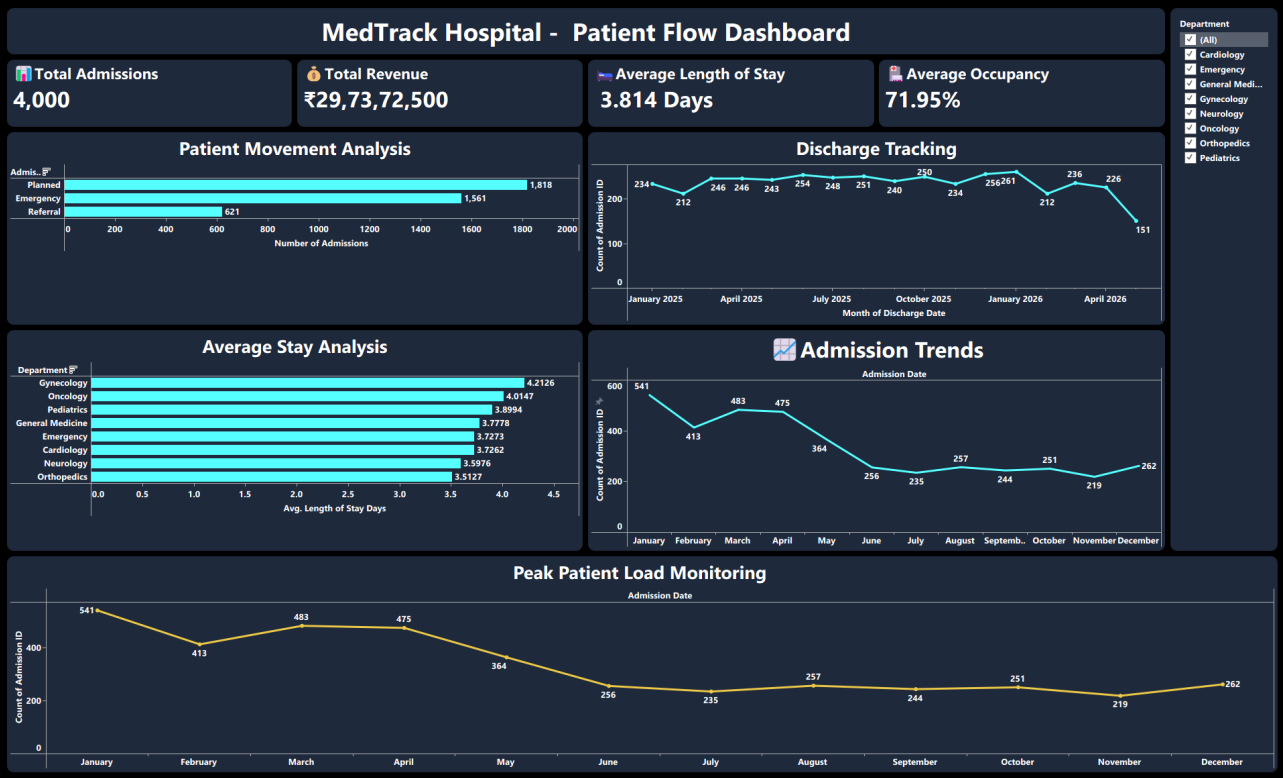
The following pages contain the final dashboard screenshots produced for the MedTrack Hospital Analytics project. They are included as visual evidence of the completed Tableau workbook and portfolio-ready dashboard suite.

Dashboard 1 — Hospital Overview & Analytics



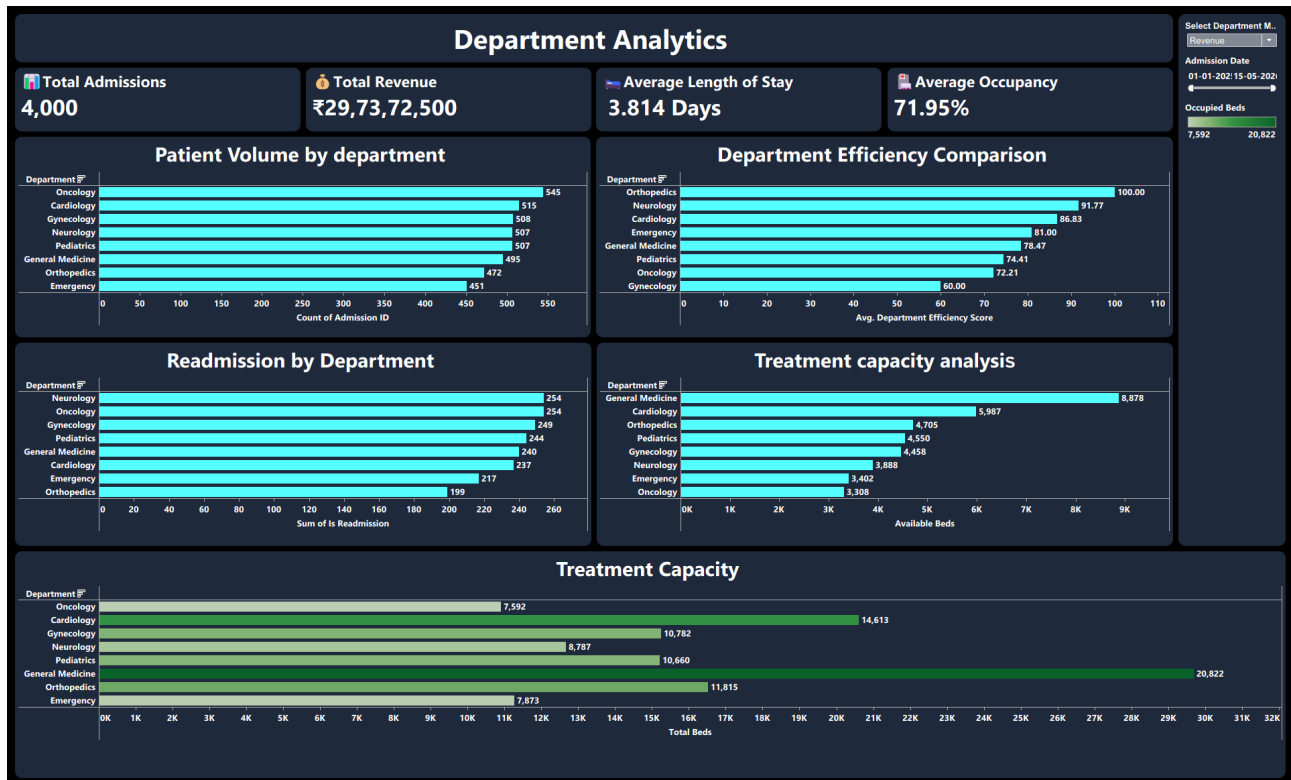
Final Tableau dashboard screenshot — included as part of the Module 8 project delivery documentation.

Dashboard 2 — Patient Flow Dashboard



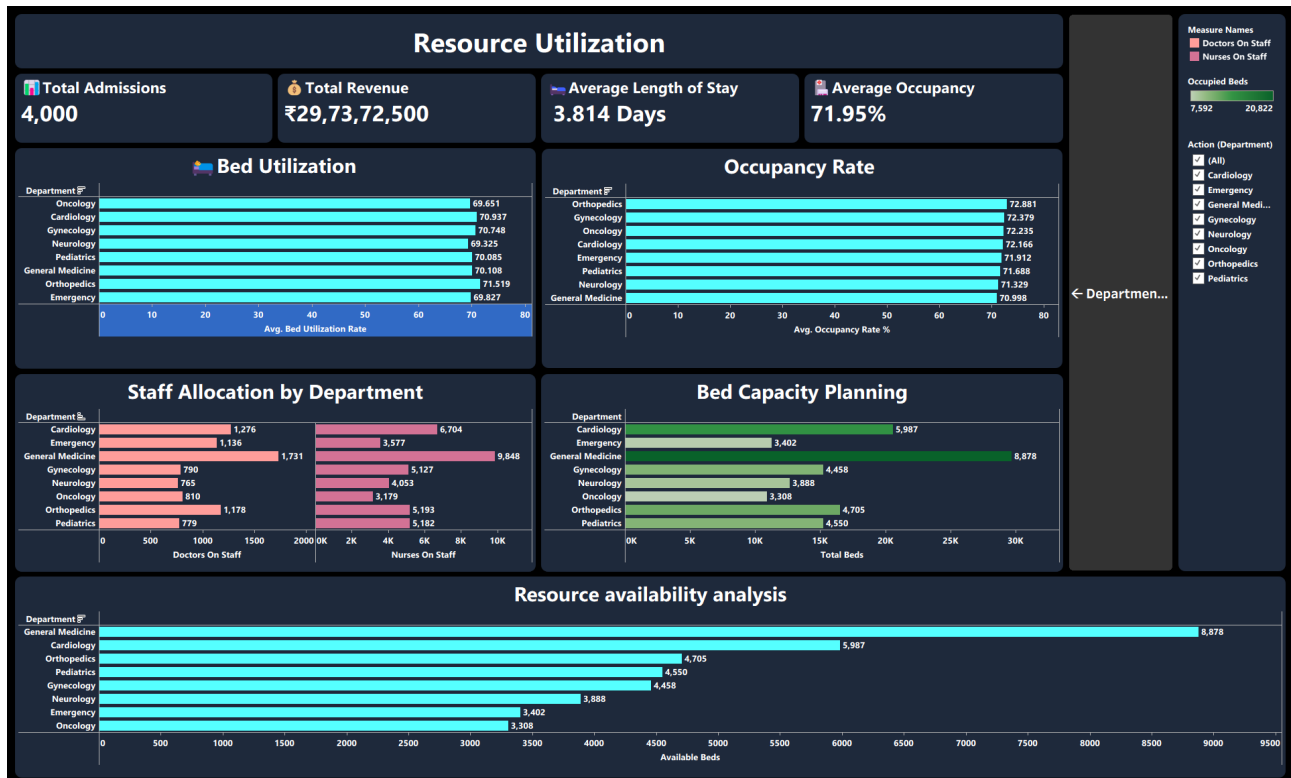
Final Tableau dashboard screenshot — included as part of the Module 8 project delivery documentation.

Dashboard 3 — Department Analytics



Final Tableau dashboard screenshot — included as part of the Module 8 project delivery documentation.

Dashboard 4 — Resource Utilization



Final Tableau dashboard screenshot — included as part of the Module 8 project delivery documentation.